

A Non-Associated MSCC Model for Simulating Structured and Destructured Clays

Bohan Jing
Beijing University of Technology
Beijing, China
jingbohan2007@163.com

Weiran Li
Beijing University of Technology
Beijing, China
liweiran@bjut.edu.cn

Qing Zhu
Beijing University of Technology
Beijing, China
ccgszq@bjut.edu.cn

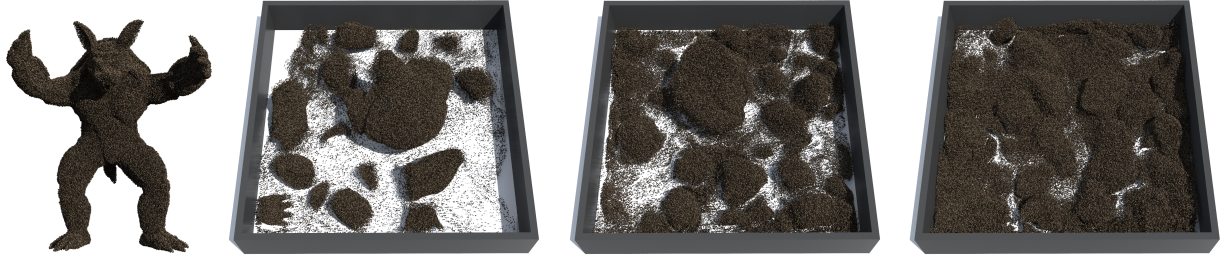


Figure 1: Clay armadillo smash. Clay armadillos are smashed on the ground under different gravity conditions.

ABSTRACT

Soil structure describes the association and arrangement of the soil granules. Structured clays, such as the cracked dry ground, exhibit characteristics of solid-like materials, while destructured clays exhibit characteristics of granular materials. The structure strength weakens with the deviatoric plastic strain accumulation, which leads to clay degradation. However, the existing graphics methods for granular materials cannot simulate this unique mechanical behavior. We present a non-associated Modified Structured Cam Clay (MSCC) model for simulating structured and destructured clays. We track the deviatoric part of the plastic Hencky strain and design two degradation functions (corresponding to yield stress and mean effective stress) to control the size of the yield surface. Through a non-associated flow rule, our method generates visually plausible results while allowing volume preservation.

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1 INTRODUCTION

Granular materials are ubiquitous in daily life and can generate impressive visual effects. Many researchers seek to animate granular material phenomena via various yield functions and flow rules. However, existing methods have limitations related to generalizing destructuration and consolidation characteristics. The Drucker-Prager-based sand simulation [Klár et al. 2016] produces convincing

behaviors of dry sand, but it lacks cohesion and cannot produce fractures while the material is undergoing hydrostatic pressure. Gaume et al. [2018] proposed the Cohesive Cam-Clay (CCC) model for simulating snow by adding cohesive terms to the Modified Cam Clay (MCC) model. Their cohesion is based on the consolidation pressure, so it cannot be degraded independently. We propose a novel degradation function to simulate structured and destructured clays, that allows us to control the increasing part of mean effective stress and simulate softening behaviors due to the void ratio.

2 METHOD

According to the Modified Structured Cam Clay (MSCC) model [Suebsuk et al. 2010], the destructuring mechanism is correlated to the deviatoric plastic strain accumulation. We adopt the Hencky strain measure, and define the deviatoric part of the plastic strain in principal space as:

$$\epsilon_d^P = \frac{1}{1+\nu'} \sqrt{\frac{1}{2} \left[(\epsilon_1^P - \epsilon_2^P)^2 + (\epsilon_2^P - \epsilon_3^P)^2 + (\epsilon_3^P - \epsilon_1^P)^2 \right]}$$

Where ν' is the effective Poisson's ratio, $\epsilon_i^P = \log(\Sigma_{ii}^P)$ and Σ_{ii}^P is the singular value of the plastic deformation gradient. We use the non-associated flow rule [Wolper et al. 2019] to ensure the projections preserve volume as much as possible. Then our non-associated MSCC yield surface can be described as:

$$y(p^{tr}, q) = q^2 + M^2(p^{tr} + p_b)(p^{tr} - p_c)$$

Where M is the slope of the critical state line (CSL) in the pq -plane, p_c is the pre-consolidation pressure, and p_b represents the increasing part of mean effective stress. p_b evolves according to:

$$p_b = \begin{cases} p_0 e^{-\epsilon_d^P} & \text{if } \epsilon_d^P \leq \epsilon_{d,f}^P \\ p_f e^{-\xi_d(\epsilon_d^P - \epsilon_{d,f}^P)} & \text{if } \epsilon_d^P > \epsilon_{d,f}^P \end{cases}$$

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Where p_0 is the initial structure strength, ξ_d is the destructuring factor, $\epsilon_{d,f}^P$ is the failure deviatoric plastic strain, and p_f is the failure structure strength. In the first stage ($\epsilon_d^P \leq \epsilon_{d,f}^P$), soil particles have high tensile and compressive strength. p_b decreases gradually due to the deviatoric plastic strain accumulation. Within the second stage ($\epsilon_d^P > \epsilon_{d,f}^P$), soil particles begin to destructure, which causes the structure strength to decrease rapidly.

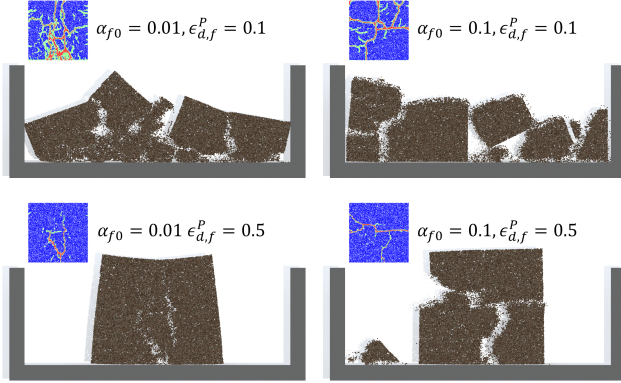


Figure 2: α_{f0} and $\epsilon_{d,f}^P$ are shown to control the fracture behavior. The colored particles show the destructuring process in our method. The blue particles are structured particles. The green particles are experiencing destructuring. And the red particles show the soil structure is completely crushed.

The hardening rule of the pre-consolidation pressure is related to the void ratio [Suebsuk et al. 2010]. But changes in void ratio leads to changes in volume, which is not visually plausible in graphics. We are devoted to simulating the effect of the void ratio with volume preservation. It is easy to prove that our yield surface has the same shape as CCC (ellipsoid), which allows us to propose an approximative model based on Fracture-Friendly Hardening [Wolper et al. 2019] to describe the mechanical behavior of the void ratio:

$$p_c = K \sinh \left(\xi_h \max(-\alpha, \alpha_f) \right)$$

Where K is the bulk modulus, ξ_h is the hardening factor, α is the Fracture-Friendly hardening state, and α_f is used to control the minimum p_c values. For structured clay, p_c cannot decrease to below the minimum value by softening. When ϵ_d^P reaches the failure value, a sudden decrease in α_f occurs to simulate the void ratio changes. We refer to the form of intrinsic compression line (ICL) and define the degradation function as a linear function. Furthermore, we assume α_f and p_b return to zero simultaneously to result in splash appearance. The degradation function of α_f is defined as:

$$\alpha_f = \begin{cases} \alpha_{f0} & \text{if } \epsilon_d^P \leq \epsilon_{d,f}^P \\ \alpha_{f0} \frac{\epsilon_d^P - \epsilon_{d,e}^P}{\epsilon_{d,f}^P - \epsilon_{d,e}^P} & \text{if } \epsilon_{d,f}^P < \epsilon_d^P \leq \epsilon_{d,e}^P \end{cases}$$

Where α_{f0} is the initial value of α_f , and $\epsilon_{d,e}^P$ is the deviatoric plastic strain when $p_b = 0.001$. We choose the Material Point Method discretization scheme and solve equations in the diagonal space.

3 RESULT AND FUTURE WORK

In Figure 1, we demonstrate the efficacy of our approach by smashing a clay armadillo with 7 million particles. When the gravity level is sufficiently low (left), most particles still preserve structure, resulting in a solid-like appearance. As the gravity level increases (middle and right), the impact leads to an increase in plastic strain which produces more destructured particles. In Figure 2, we show the efficacy of our parameters. In Figure 3, a high-velocity stone hits a clay castle with 6 million particles. The impacted part of the castle is undergoing destructuration, while the bottom part remains structural.

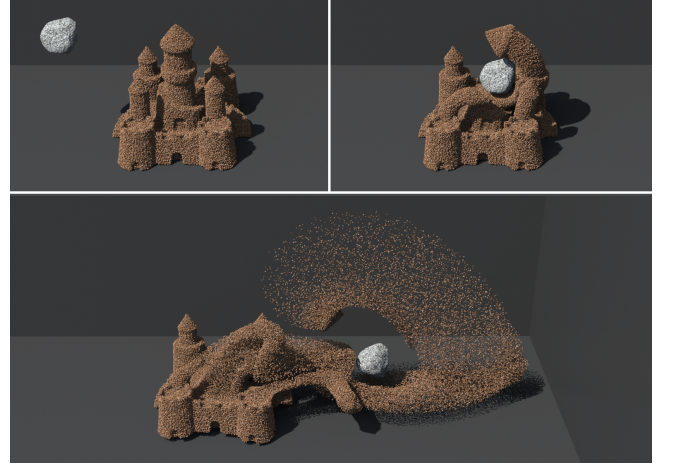


Figure 3: A clay castle is hit by a stone, producing large amounts of debris.

Although our method can show visually appealing results of structured and destructured clays, it does have some limitations. The definition of α_f is a simplification of physical model, so it cannot be determined from physics experiments. For future work, we plan to develop more suitable hardening rules based on physics and simulate the coupling mechanisms of soil and water.

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The castle model courtesy of assetfactory, available under a Creative Commons Attribution 4.0 Licence at <https://skfb.ly/ow7Q6>.

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